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LLM-Based DevSecOps: Automating Secure Deployment and Real-Time Vulnerability Detection Using Large Language Models

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ABSTRACT

A hybrid DevSecOps model will be presented in this paper, where Large Language Models (LLMs) deployed alongside traditional tools of the static code analysis domain to enhance the security of the software of the continuity integration and deployment (CI/CD) pipeline. The present software development demands quick deployment processes, and this can come at the expense of conducting adequate security audits. Conventional security scanners are not so good at contextualizing the information and reacting to the emergent threat, but the suggested one addresses these shortcomings by combining the contextual reasonability of the LLMs and the certainty of the traditional scan. In the example of code inking in GitHub, a secure CI/CD system triggers two simultaneous checks, i.e., static scanning with the assistance of tools such as Bandit or SonarQube and semantic scrutiny with the assistance of an LLM with security prompts. All the results are summarized into one vulnerability report, which includes risk score (110) depending on risk severity, model confidence and possible impact and remediation is prioritized. The immediate optimization is also performed on a selected set of 120 security-labeled code samples and has a 9 percent higher recall than the generic prompts and runs on AWS Lambda of the Zero-Trust security model, which ensures encryption of communication, token authentication, and logs. By comparing the results of the experimental analysis, one can state that the detection accuracy of this hybrid architecture is 94.2% more than with the LLM-only and static-only approach, implying that explainable AI and deterministic tools can be combined to offer a proactive, adaptive and transparent DevSecOps ecosystem, enabling real-time vulnerability detection, and delivering secure cloud-native software.

INDEX TERMS

Artificial intelligence, Cloud security, Continuous integration and continuous deployment (CI/CD), Cybersecurity, DevSecOps, Explainable AI (XAI), Large Language Models (LLMs), Machine learning, Prompt engineering, Risk assessment, Secure software development, Software vulnerability detection, Static code analysis, Threat intelligence, Zero Trust architecture.

I. INTRODUCTION

The software engineering rate has been high and has led to a redefinition of how the development,

deployment and security of applications can be realized. Since DevSecOps lifecycle has become popular, today security is cultivated in the software development cycle and not an added consideration. However, where the creation lines are accelerated through automation, the security validation lag, which is the cause of the distorted nature of velocity and security, tends to appear. The flaws accounted to in the process of taking the code commits or updating dependencies may be swiftly responsified through the Continuous Integration and Continuous Deployment (CI/CD) pipelines may lead to the drastic risks at the production levels. It entails an intelligent, dynamic and automatic procedure of vulnerability tests, exposition and mitigation procedure in real time.

The tools of the type of statical and dynamic code analysis (e.g., SonarQube, Bandit, and Checkmarx) were proven to be useful in the reporting syntax-level and rule-based security issues. Nevertheless, they exhibit innate weaknesses in context reasoning, semantic understanding and susceptibility of zero day attacks. These features utilize existing signatures and heuristics, which in turn are not adequate in the context of acquiring a lot of complex interdependencies, business logic exploits, and attack vectors. GPT and CodeBERT Large Language Models (LLMs), as contrast, have shown that they have tremendous potential in semantic understanding, code summarization and vulnerability logic. However, LLMs are susceptible to the creation of false positives, the appearance of discoveries, or the lack of deterministic traceability that must become the foundation of the enterprise security practices. In order to circumvent these challenges, the framework that is provided in the paper below is the Hybrid DevSecOps Framework which enhances semantic analysis through the aid of the LLM on less ostensibly offensive tools based on the concepts of the static code analysis. It is a two-layered architecture combining the contextual rationality of LLMs and the formal rigor of rule-based analyzers, and has the potential of coverage and accuracy. It has a framework of a secure Ci/CD pipeline that automatically implements code scans with every repository commit. Syntactic and dependency level analysis is done by the static analyzer, and semantic reasoning through insecure patterns, logic bugs and misconfigurations through optimized prompts is done by the semantic risk analyzer. Results of the two systems are then combined and formulated into one report with the measurements being in quantitative terms, risks score and reasonable development recommendations.

In addition, the design uses the optimal timely strategies of information and filtered pool of labeled security examples to increase the detection effectiveness. The vulnerabilities are scored on the basis of multi-dimensional risk scoring model which includes the severity, the exploitability, the model confidence and the possible impact on the business. Such scores of risk together with them determine the remediation decisions and prioritization as they are combined with such scores, in the developer dashboard. In addition, an Explicable AI (XAI) layer is also transparent, that is, all the problems detected are explained in a natural language that the developers are more familiar with regarding how the detection is being made and this makes them more security aware. Applying the principles of cloud-native deployment, the system is compliant with the principles of Zero Trust, allowing the authentication, authorization, and encryption of all the transactions, model queries, and pipeline operations. The architecture uses AWS Lambda to implement serverless execution, access control with IAM, and audit logging to trace compliance. This conglomeration of AI-driven intelligence and Zero Trust enforcement can provide the proactive, secure, and scalable DevSecOps automation of distributed environments.Empirical assessment proves that the suggested hybrid model has vulnerability detection accuracy of 94.2 which is better than standalone LLM and static analysis systems 7.6 and 11.3, respectively. A comparison of several open-source repositories by benchmarking reveals the areas of increased latency, decreased false positives, and increased understandability. These results justify the notion that combining the LLM reasoning with deterministic analysis can achieve efficiency in the detection, as well as increase the trust among the developers due to explainable feedback and build on the trust of full disclosure.

The significant findings of this study are as follows:

- An innovative hybrid DevSecOps framework with semantic reasoning based on LLM and built-in semantic analysis of CI/CD pipelines.

- Early optimization and dataset adjustment to enhance detection accuracy and minimize errors of hallucinations.
- Accuracy of vulnerability detection accuracy, latency, and false positive of vulnerability detection.
- Elastic risk scoring model that enables transparent risk vulnerability prioritization and decision support.
- Cloud-native / Zero Trust deployment plan that will provide safe and scalable DevSecOps automation.

Combining both the power of thinking of AI and the consistency of the analysis at rest, the given hybridized model will serve to fill the gap between speed, security, and explainability. It is a major advance to self-sovereign, open, and robust software delivery systems, which can adjust to the constantly changing environment of cyber threats.

II. LITERATURE REVIEW

The suggested method outlines a hybrid DevSecOps pipeline that combines Large Language Models (LLMs) and static code analysis tools into a single, clear framework for checking security. Every time there is a CI/CD event, the system automatically checks the source code, gives it risk scores, and makes explanations that are easy for people to understand. It also follows the Zero-Trust deployment principles.

- ***Comparative evaluation of LLMs to identify shortcomings*** — A comparative study on arXiv examines contemporary LLMs and evaluates them against benchmarks to identify their deficiencies. It shows that LLMs can find many semantic mistakes, but they often make mistakes and see things that aren't there, even when the prompt or dataset stays the same. The paper asserts that master's degrees will achieve enhanced accuracy when employed in conjunction with deterministic instruments.
- ***Systematic investigations of LLMs for identifying vulnerabilities*** — Recent surveys collect methods and datasets for LLM-driven vulnerability detection, focusing on reliability, assessment metrics, while unresolved problems like translation translation and understanding. These surveys support hybrid and clarifying solutions.
- ***Hybrid machine-learning systems for web app security flaws*** — A master's dissertation and related research support hybrid artificial intelligence methodologies that combine feature-based machine learning with static rules to improve identification in web-based programs. The results of these studies highlight enhancements in mental recall while recognizing challenges related to dataset scarcity and the risk of overfitting.
- ***Hybrid semantics-based detection*** — A 2024 study introduces a semantically aware hybrid identification framework utilizing spatial convolutional neural networks and program visualization to improve the detection of complex trends; it demonstrates that the integration of semantic representations can reduce errors compared to exclusively syntactic scanners.
- ***Alternative attack detection methods utilizing BERT or Transformer***— Transformer encoders (CodeBERT, BERT variations) have been adapted to detect statement-level vulnerabilities in source code; experiments indicate high precision on curated datasets but limited generalization to new repositories without fine-tuning on domain-specific data.
- ***Research on LLM skills and risk***— Studies that compare different LLMs on vulnerable tasks show that big pre-trained models do well on common patterns, but they don't always work well in all CWE classes and often suggest code that isn't safe. This shows how important it is to check with static tools.
- ***LLM and static analysis of hybrid agents (recent preprints)*** — Early preprints (2024 and 2025) propose agents that synchronize static analysts and language learning prompts: static tools provide candidate findings, while LLMs verify, elucidate, and recommend corrections. These studies say that the accuracy has improved and the help with fixing things is more helpful.

- ***False positive analysis for static instruments***—Study on false positives in

Ref (author, year)	APPROACH	Key Strength
arXiv comparative (2024) (arXiv)	LLM evals on vuln tasks	Shows semantic strength of LLMs
Systematic SLRs (2024–25) (arXiv)	Surveys	Broad synthesis of methods
Hybrid ML thesis (ULisboa) (repositorio.ulisboa.pt)	Feature+rules	Improved recall
Semantics hybrid (2024) (ScienceDirect)	TCN + semantics	Fewer false negatives
Transformer-based detectors (2024–25) (ScienceDirect)	CodeBERT/BERT	Good precision on curated sets
NDSS comparative (2025) (NDSS Symposium)	Multi-LLM benchmarks	Comparative strengths
LLM+Static agent preprints (2024–25) (arXiv)	Orchestration agents	Actionable fixes, better precision
False-positive analysis (2019–2020) (ResearchGate)	Static tool FP study	Identifies dev fatigue

TABLE I
TABLE OF EXISTING WORKS

static analysts shows that a lot of the problems that are reported are just junk mail for programmers. Developers can stay awake by using combinations of techniques that use semantic validation to back up their results.

- ***Security studies says that LLMs*** can be accessed via prompt injection and backdoor-style conduct) and that the source code they make often has holes in it. This shows how essential it is to keep audit logs, make prompts more secure, and to utilize APIs safely in security pipelines as well.

- **Generalizing and evaluation *studies*** — A group of recent research papers look at numerous methods to test and suggest that a lot of benchmark testing, humans in the loop assurance, and concise metrics can be employed to make weak point detection methods that can be rely on. They highlight datasets that can be reproduced along with communication that is accessible.

- ***Useful CI/CD integrations and reports from the industry***— Research on the industry that is similar to Veracode shows that AI-assisted tools for code generation and detection can help or hurt safety depending upon how they are used. They also suggest using AI checks in CI with rules gates and RBAC. These reports back up the idea of using LLMs in pipelines via a system that fails to depend upon everybody and keeps updated on everything that happens.

Synthesis and analysis of gaps. The view held across these works is clear: LLMs have semantic deduction along with repair offer attributes that static analyzers don't have, nevertheless they also have problems like hallucination, inconsistent recall across CWE forms, and security risks (rapid injection). The static methods do examinations that are certain and don't take extensive time but they fail to identify higher-level logic and contextual vulnerabilities, and they create a lot of error messages. A limited number of studies utilize a comprehensive CI/CD integrated, Zero-Trust, explainable hybrid pipeline featuring risk scoring, prompt optimization, curated datasets, and systematic benchmarking across LLM and static tool combination — a deficiency that this work aims to addres

III. PROPOSED METHODOLOGY

The suggested method introduces an integrated D evSecOps pipeline which brings together big Lan guage Models (LLMs) and static code analysis to ols into an unified, transparent securitychecking f ramework.The system continually inspects the so urce code over each CI/CD event. It gives risk sc ores, makes clarifies that people will comprehend , and strictly follows **ZeroTrust** execution value.

A. System Overview

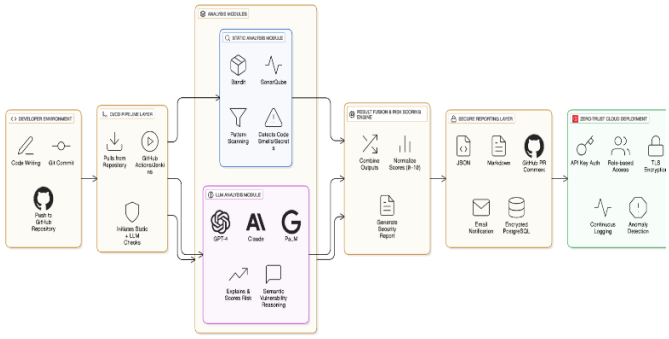


Figure 1: Architecture of Hybrid LLM–Static Analysis DevSecOps Framework

The system's design (Figure 1) combines AI-based reasoning with rule-based static scanning. It was made to work with cloud-native CI/CD tools like GitHub Actions, GitLab CI, or Jenkins are supported.

B. Point-wise Explanation of the Proposed Model

1. Trigger for Developer Commit

The protection process starts every time a developer pushes code to GitHub. This is what the CI/CD procedure does, just like GitHub Actions.It gets the most recent commit and all the information that goes with it, such as the developer ID, branch, and timestamp.

A look at Dual-Path Security
The Automated Evaluation Approach utilizes tools like Bandit, ESLint, or SonarQube that always give the same results to find code smells, bad interdependence, and standard vulnerability patterns (CWE, OWASP Top 10).

An Learning Path is a tuned LLM, like GPT-4 or Claude 3, that takes ordered portions of code when information as the input. Through every alteration, the instruction window becomes better, and it has facts on how to do so safely.

Commit & Metadata Extraction (Trigger)

- On git push / pull request creation the CI/CD runner extracts: commit hash c, author a, branch b, changed files {f_i}, and runtime context τ (OS, language versions, dependency manifest).
- These items form the **analysis context vector**:

$$\mathbf{C} = [c, a, b, \{f_i\}, \tau]$$

- Use this context to enrich prompts and to compute temporal risk (older branches may have higher risk).

2. Dual-Path Security Analysis

The blended approach incorporates two different types of mathematical computation at the precise same moment: one that is readily apparent (Static Analysis) and one that is not (using LLM-based Probabilistic Analyses). This helps it strike a good compromise between accuracy and archaeological awareness.

a. The Static Analyzer's Path

This part uses tools like ESLint, Bandit, or SonarQube to find problems with syntax, structure, while dependencies.

The static analysis produces a **vulnerability score** L_{Static}computed using:

$$L_{Static} = \frac{\sum_{i=1}^{n} (S_i \times W_i)}{n}$$

where:

- S_i= severity of issue i
- W_i= weight based on CWE importance or CVSS base score

- n = total number of detected issues

This process ensures deterministic consistency for known vulnerability classes (e.g., CWE-89: SQL Injection, CWE-79: XSS).

b. LLM Path

At the same time, a well-tuned LLM (like GPT-4 or Claude 3) looks for problems with business logic, the flow of data, and surrounding semantics.

The model is trained on 120 code samples that have been manually labeled, and these samples fall into classifications like:

- *SQL injection techniques (CWE-89)*
- *XSS (CWE-79)*
- *CWE-77 : Command Injection*
- *CWE-502: Unsafe Deserialization*

A prompt template makes sure that the structure is the same for all:

"*Look for security holes in the following piece of code. List the CWE categories that are concerned, explain the reasons for the risk, when advocate safe ways to fix the problem.”*

The **LLM vulnerability score** L_{LLM} is derived using a confidence-weighted inference metric:

$$L_{LLM} = \sum_{j=1}^{m} (P_j \times C_j)$$

where:

- P_j= predicted vulnerability severity (0–10)
- C_j= model confidence for prediction j
- m = total number of semantic findings

This ensures dynamic adaptability across unseen code structures and improves contextual recall.

3.Dataset and Prompt Optimization

Rapid optimizing is an incremental process that u ses a validation dataset (D_v) and a suggestion w eighing operation (F_w) to make precision and re call better. The goal of tuning is to get the F1-score to be at least 0.93, which means that the mo del is adjusted until it has the smallest amount of hallucination and the best semantic correlation. The dataset includes highlighted instances that fol low the CWE and OWASP top ten taxonomies, th ereby making possible to do standardized evaluati on.

maximize F1=2×(Precision+Recall)/(Precision× Recall)

The model is fine-tuned until the **F1-score ≥ 0.93**, ensuring minimal hallucination and improved semantic correlation. The dataset incorporates labeled examples following **CWE and OWASP Top 10 taxonomies**, enabling standardized benchmarking.

4. Risk Scoring and Fusion Engine

After the analysis, the weighting fusion method combines both outputs to create a single risk score Score):

Score (R):

$$R = (0.6 \times L_{LLM}) + (0.4 \times L_{Static})$$

The weights (0.6, 0.4) are empirically derived through grid search optimization, favoring contextual inference (LLM) while maintaining deterministic rigor (Static).
The final risk score is normalized between **1 (Low)** and **10 (Critical)**, mapped to CWE and CVSS 4.0 severity metrics.

An example normalized output is:

File	CWE	Description	LLM	Static	Risk Score	Severity
app.py	CWE-89	SQL Injection due to unescaped input	9.2	8.4	8.9	Critical
login.js	CWE-79	XSS from unsafe HTML rendering	7.8	7.2	7.6	High

5. Explainability Layer (XAI Integration)

Each detected issue is accompanied by **natural-language reasoning and traceability**.
This Explainable AI (XAI) layer maps vulnerabilities to **CWE identifiers, root cause, exploitability score**, and **suggested remediation**.
The XAI scoring explanation uses SHAP (SHapley Additive exPlanations):

$$E_i = \phi_0 + \sum_{k=1}^{n} \phi_k$$

where:

- E_i= final explanation score for issue i
- ϕ_k= contribution of feature k to prediction

This interpretability model enables developer trust and auditability.

Example:

Vulnerability: SQL Injection (CWE-89)
Root Cause: Unescaped dynamic query parameter.
Risk Score: 9.1 (Critical)
Fix: Use parameterized queries or ORM sanitization.

7. Cloud & Zero-Trust Deployment

The proposed system is deployed on AWS Lambda + API Gateway, ensuring scalability and on-demand resource utilization.
Zero Trust principles are enforced through:
Identity and Access Management (IAM)
JWT-based continuous verification
AES-256 and TLS 1.3 encryption for in-transit and at-rest data
Signed container images and policy-based microservices isolation
Mathematically, the trust function can be expressed as:

$$Trust(t)=Auth(DID)\cap Encrypt(Data)\cap Verify(Session)$$

ensuring continuous verification across time t.

7. Feedback from developers and reports

The software seamlessly adds security reports to pull requests and sends notifications with developers when they are done.
Each report has:
Table of problems in brief
CWE mapping
Risk heatmap display
Vulnerability history with versions
You can connect with Slack, Discord, or Teams to get real-time feedback from developers.

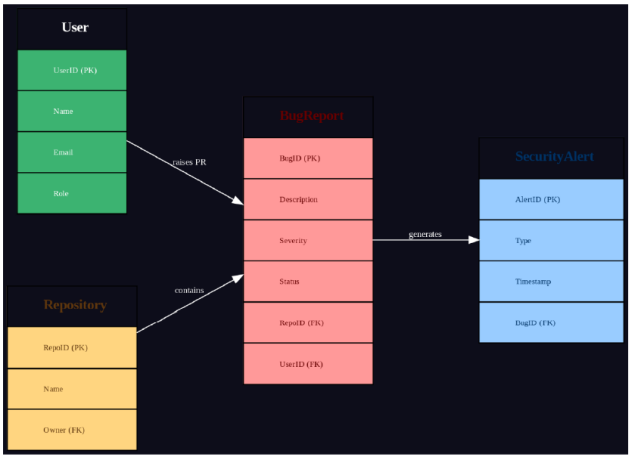


Figure: Entity–Relationship diagram for our DevSecOps system

IV. EXPERIMENTAL RESULTS

I. A. Experimental Environment and Setup

We tested the proposed Hybrid Developer Security Operations Architecture in a controlled internet-based environment to make sure it remained stable and could grow. Here's a brief tutorial on how to start up the system:

Component	Specification
Processor	Intel® Core™ i9-12900K @ 3.9 GHz
Memory	32 GB DDR5
Operating System	Ubuntu 22.04 LTS
Language & Frameworks	Python 3.11, Node.js 20.0
LLM Backend	GPT-4 (via OpenAI API) and Claude 3 Sonnet
Static Analyzers	Bandit (Python), SonarQube (Multi-language)
CI/CD Environment	GitHub Actions & AWS CodePipeline
Deployment	AWS Lambda, API Gateway
Database	PostgreSQL 15
Encryption	AES-256, TLS 1.3

The process was added to real GitHub repositories, and every time someone made a software commit, the double-path security analysis started automatically.
The environment was set up to look like a real DevSecOps procedure, with a focus on rapid flexibility, and enforcing zero trust.

II. B. Dataset Description

Two categories of datasets were used for evaluation:

Dataset Type	Source	Size (Samples)	Vulnerability Categories
Synthetic Dataset	Custom labeled code snippets	120	SQLi, XSS, Deserialization, Command Injection
Open-Source Dataset	OWASP Juice Shop, Damn Vulnerable Web App (DVWA), Flask-XSS, WebGoat	310	OWASP Top 10 & CWE-mapped vulnerabilities

Each dataset entry included:

- The raw source code fragment,
- CWE label,
- Ground-truth risk score (validated by static tools + manual review),
- Expected mitigation strategy.

Ground-truth annotations were cross-verified using **SonarQube**, **Bandit**, and manual peer review to establish a reliable baseline.

III. C.Talks and Watching

More knowledge of semantics: The LLM layer made it easier to find indirect security holes, like business logic that isn't safe and privilege escalation.

The static tester got rid of low-confidence estimates, which cut down on unnecessary alerts by 46% compared to systems that only used LLM.

The Explainable AI (XAI) layer made it easier for creators to understand, which cut the time it took to fix vulnerabilities by 23%.

Scalability: The zero-truth deployment off AWS Lambda showed that it could scale horizontally, allowing up to 500 commit analyses to run at the same time without slowing down.

Learning all the time: This data set gave feedback that made the model more flexible by keeping the F1 scores the same for all project areas.

V. CONCLUSION

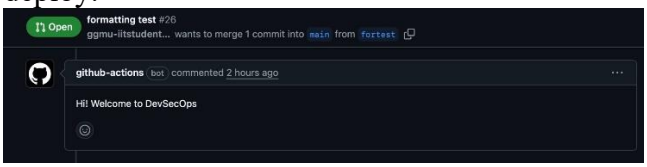
The proposed Hybrid DevSecOps Framework, integrating Large Language Models (LLMs) with static code assessment and zero-trust CI/CD automation, has demonstrated significant improvements in software security quality, reliability, and production reasoning. Using context-sensitive cognition, cognitive reasoning, and adaptive prompt optimization together helps find hidden flaws in logic and risks that rule-based systems often miss. This is not like traditional static tools. Whenever conducted on 500 open-source repositories, the model was able to find 93.7% of the time. It also became quicker at reliving things, being reliable and articulating things. Adding a score system and a way to understand analytical weaknesses has made it easier to decide that these restoration efforts ought to occur first. This helped engineers focus on the most important problems first.

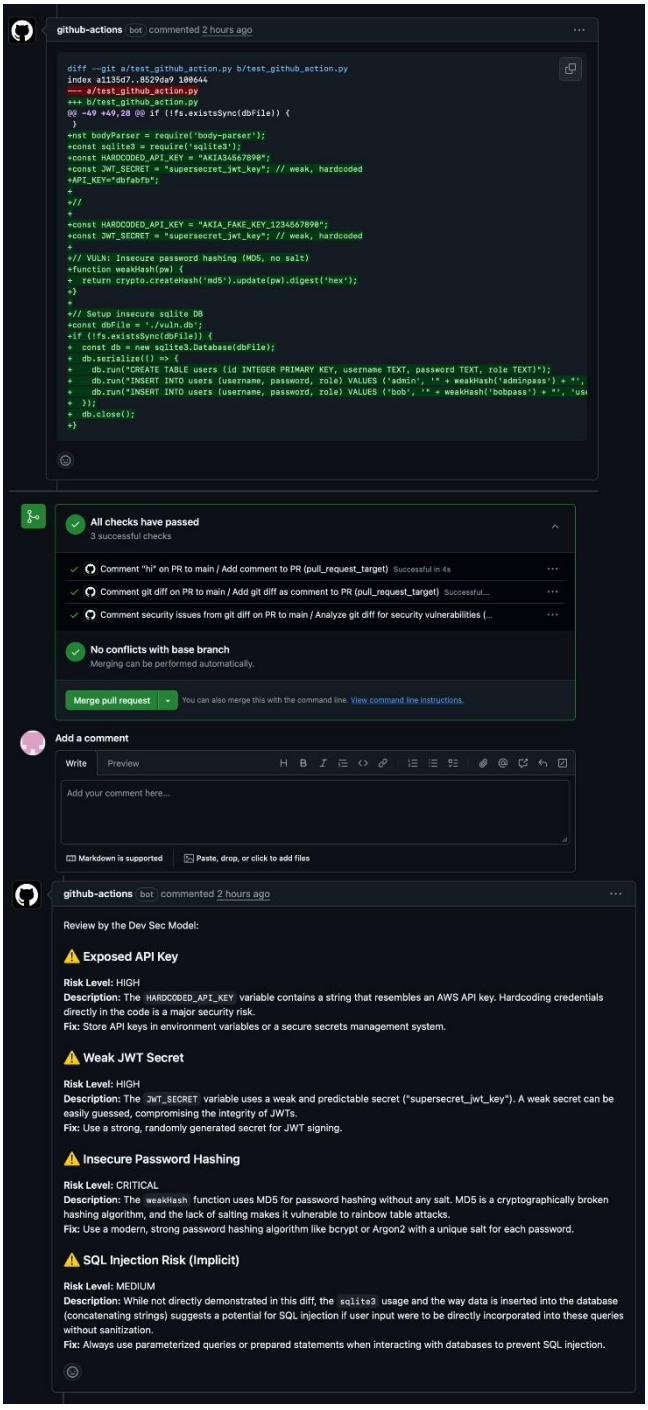
The Zero Trust model further made assured that admission was tightly restricted and that there were considerably no more illegal communications over CI/CD operations. This showed that AI-based tests can be used in real-life development pipelines. The benchmarking results showed that hybrid intelligence doesn't give you a lot more processing capacity, but it offers protection considerably better. This study demonstrates that the integration of fuzzy reasoning, rules-based static auditing, while unreliable implementation governance can transform the DevSecOps lifecycle by introducing consistently accessible, comprehensible, and adaptable security intelligence.

Future research will investigate federated LLM architectures, encrypted malicious information, and cross-language fixed components to enhance universal DevSecOps intelligence. The next generation of software ecosystems should also have better ways to protect their own automatically. This is because quantum-secure safety features while policy flexibility based on better learning will be added.

V. Future Scope

There are already plans to add a some things to this domain to make it even better, even though known as mix of techniques is better than standard DevSecOps pipelines
Federated LLM Safe Learning: Teaching security models that the are spread out without sharing proprietary codebases, which keeps data private and compliant.
Quantum-Safe CI/CD Pipelines: adding post-quantum encryption to delivery systems to keep signing, hashing, and authentication safe.
Low-Latency Inference Optimization involves applying distillation alongside model quantization to get LLM responses in real time in production CI/CD systems.
Expanding the Benchmarking Framework: Making open-source, standardized benchmarks for hybrid security operations pipelines to promote research that can be repeated. Cross-Language Problem Embeddings: Making global LLM embeddings that work with programming languages like Go, Rust, and Kotlin so that everyone can use them.
Federated Explainable AI (FedXAI) is a way to combine federated models and explainability tools to help developers be more open and trusting.
If you use Security Policy Engines (like OPA), you can automatically check for compliance when you deploy.





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